Flywheel Rotor Fatigue Life & Dynamic Reliability Assessment

for AI Mega-Cluster DC-Bus Systems

Technical Engineering Report
Date: 2026-05-21
Material: 25Cr2Ni4MoV | Max Speed: 20,000 rpm | 800V DC Bus | SiC DC/DC + SST

Abstract — This report presents a comprehensive fatigue life and dynamic reliability assessment of a 25Cr2Ni4MoV forged steel flywheel rotor (φ430 mm, 20,000 rpm) for AI mega-cluster GPU data center applications. Unlike traditional UPS scenarios, AI data centers induce broadband random electromagnetic torque excitation (10 Hz–10 kHz) via GPU Compute Spikes. A full-chain coupling fatigue analysis framework is established. Key results demonstrate infinite fatigue initiation life (safety margin $> 11.8\times$) and bounded crack propagation ($\Delta K = 1.38$ MPa $\cdot\, m^{\frac{1}{2}}$ vs. threshold 5–8), yielding a total design life exceeding 20 years under standard NDT conditions.

Keywords — flywheel energy storage, AI data center, rotor fatigue, 25Cr2Ni4MoV, dynamic reliability, broadband random excitation

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1 Introduction

1.1 Background

The rapid scaling of AI computing infrastructure has introduced unprecedented challenges in power delivery and backup energy storage. Modern GPU mega-clusters, comprising 10,000+ accelerators, exhibit highly dynamic power consumption profiles characterized by broadband fluctuations spanning 10 Hz to 10 kHz [1]“. These fluctuations arise from computation synchronization events (All-Reduce, gradient checkpointing), inference request burstiness, and thermal management cycles, creating a fundamentally different load regime compared to traditional data center UPS applications.

1.2 Problem Statement

Flywheel energy storage systems (FESS) coupled to 800V DC buses through SiC DC/DC converters and solid-state transformers (SST) have emerged as a promising solution for AI data center power quality management and short-duration backup. However, the existing fatigue life assessment framework for flywheel rotors is predicated on narrowband grid-frequency excitation (0.01–0.5 Hz), which is fundamentally inadequate for AI data center conditions.

The critical engineering questions are:

1. **Fatigue initiation**: Under broadband random torque excitation, does the rotor’s fatigue life remain infinite as in traditional UPS scenarios?
2. **Crack propagation**: What is the crack growth life from manufacturing defects under AI data center load spectra?
3. **Power scalability**: Does the fatigue life conclusion hold across the 500 kW–3200 kW power range?

1.3 Research Scope

This report establishes a full-chain coupling fatigue analysis framework covering:

- **Section 2**: System modeling – GPU power spectral density, SiC electromagnetic coupling
- **Section 3**: Torsional vibration and Campbell analysis
- **Section 4**: Steady-state and dynamic stress decomposition
- **Section 5**: PSD-based broadband random fatigue assessment
- **Section 6**: Paris-law crack propagation analysis
- **Section 7**: Quantitative comparison with traditional UPS methodology
- **Section 8**: Engineering conclusions and design recommendations

1.4 Methodology Overview

The analysis methodology integrates multiple engineering domains:

- **Power electronics**: SiC DC/DC converter ripple modeling with space vector PWM harmonics
- **Structural dynamics**: Continuum torsional vibration with Rayleigh-Ritz modal truncation
- **Fatigue mechanics**: PSD-based Dirlik broadband rainflow with Basquin S-N curve
- **Fracture mechanics**: Paris-Erdogan law with random load spectral integration

All calculations are based on the 25Cr2Ni4MoV forged steel rotor ($$430 mm, 20,000 rpm) from Honghui Energy’s commercial FESS product, with material properties validated against ASTM A471 and GB/T 3310-2019 standards.

2 System Modeling

2.1 Flywheel Rotor Physical Parameters

Table 1 summarizes the core physical parameters of the Honghui Energy 25Cr2Ni4MoV forged steel rotor.

2.2 AI GPU Cluster Power Spectrum Model

2.2.1 GPU Load Temporal Hierarchy

The defining characteristic of AI data center loads is their multiscale temporal structure. Define the instantaneous GPU cluster bus power as:

$$P_{GPU}(t) = P_{base} + \sum_{i=1}^{N} \Delta P_i \cdot \mathrm{Rect}_{[t_i, t_i+\tau_i]}(t) + P_{noise}(t) \tag{1}$$

where P_{base} is the baseline power, ΔP_i is the amplitude of the i-th power pulse, and $P_{noise}(t)$ represents residual stochastic components.

Event	Timescale	Power Change	Mechanism
All-Reduce sync	100 μs–10 ms	+30%–+80%	Gradient aggregation
Checkpoint I/O	10–100 ms	+20%–+50%	PCIe/NVLink burst
Forward/Backward	1–100 ms	±10%–±30%	Pipeline alternation
Batch transition	10–200 ms	−40%—80%	Pipeline drain
Inference spike	1 μs–1 ms	+5%–+30%	Request scaling

For mega-clusters, these fluctuations superimpose rather than cancel. Field measurements show that normalized spectral density of total cluster power exceeds −20 dB across 10 Hz–1 kHz [1], meta-isca-2023.

2.2.2 GPU Power PSD Model

Define the normalized power spectral density of a single GPU:

$$S_{GPU}(f) = \frac{P_{pk}^2}{2\pi f_c} \cdot \frac{1}{1 + (f/f_c)^2} + \sum_{k=1}^{N} A_k \cdot \delta(f - f_k) + W(f) \tag{2}$$

where: - **Term 1**: Lorentzian continuum, cutoff $f_c \approx 50$ Hz - **Term 2**: Discrete line spectrum at computed rhythms f_k - **Term 3**: White noise floor $W(f) = \sigma_{noise}^2$

For the mega-cluster, the equivalent PSD becomes:

$$S_{cluster}(f) = N_{GPU} \cdot S_{GPU}(f) \cdot \Gamma(f) \tag{3}$$

where $\Gamma(f)$ is the coherence function, $\Gamma \approx 0.3$–0.6 at low frequencies (< 50 Hz) and $\Gamma \to N_{GPU}^{-1}$ at high frequencies.

3 Electromagnetic Torque and Torsional Vibration

3.1 SiC DC/DC Coupling Path

The flywheel system interfaces with the 800V DC bus via a SiC MOSFET DC/DC converter. The SiC switching frequency $f_{sw} = 20$–50 kHz introduces high-frequency torque ripple that must be modeled.

The total electromagnetic torque is expressed as:

$$T_e(t) = T_{e0} + \Delta T_{e,load}(t) + \Delta T_{e,ripple}(t) \tag{4}$$

3.1.1 Load-Coupled Torque

GPU power fluctuations transmitted through the DC/DC converter to the flywheel motor produce:

$$\Delta T_{e,load}(t) = \frac{1}{\omega_{FW}} \cdot \frac{P_{DC,bus}(t)}{\eta_{DC/DC}} \approx \frac{P_{GPU}(t)}{\omega_{FW} \cdot \eta_{DC/DC}} \tag{5}$$

where ω_{FW} is the flywheel angular velocity and $\eta_{DC/DC} \approx 97.5\%$ is the SiC converter efficiency.

3.1.2 PWM Ripple Torque

The PWM modulation introduces current harmonics that generate ripple torque:

$$\Delta T_{e,ripple}(t) = \frac{3}{2} p \left[\Psi_{PM} \cdot \tilde{i}_q(t) + \left(L_d - L_q \right) \cdot \tilde{i}_d(t) \cdot \tilde{i}_q(t) \right] \tag{6}$$

For a 500 kW PMSM flywheel motor, $\left| \Delta T_{e,ripple} \right|_{rms} \approx 0.5$–$2.5$ N·m.

3.1.3 Synthesized Torque Spectrum

The combined torque perturbation PSD is:

$$S_{Te}(f) = \frac{1}{\omega_{FW}^2 \eta^2} \cdot S_{GPU}(f) \cdot \left| H_{DC/DC}(f) \right|^2 + \sum_{m,n} \delta(f - (m f_{sw} \pm n f_e)) \cdot \Gamma_{ripple}^2 \tag{7}$$

3.2 Torsional Vibration Model

3.2.1 Continuum Torsion Equation

The rotor is modeled as a continuous torsional system:

$$GJ(x) \frac{\partial^2 \theta(x,t)}{\partial x^2} - \rho J_p(x) \frac{\partial^2 \theta(x,t)}{\partial t^2} - c(x) \frac{\partial \theta(x,t)}{\partial t} = -T_e(t) \cdot \delta(x - x_e) \tag{8}$$

where $\theta(x,t)$ is the angular displacement, G the shear modulus (79.5 GPa for 25Cr2Ni4MoV), and x_e the torque application point.

3.2.2 Modal Truncation

Discretization via Rayleigh-Ritz yields the N-DOF system:

$$\boldsymbol{J}\boldsymbol{\theta}^{(})(t) + \boldsymbol{C}\dot{\boldsymbol{\theta}}(t) + \boldsymbol{K}\boldsymbol{\theta}(t) = \boldsymbol{T}_e(t) \tag{9}$$

3.2.3 Natural Frequencies

Mode	f_n (Hz)	Description
1st	18–35	1st torsion
2nd	120–200	2nd torsion (bending coupling)
3rd	350–500	Higher torsion
4th+	> 800	Local modes

3.3 Campbell Diagram Analysis

Traditional UPS flywheel excitation is limited to 1×/2× rotational frequency. AI data center flywheels introduce broadband non-synchronous excitation:

- GPU PSD continuum (10 Hz–1 kHz)
- PWM sidebands at $f_{sw} \pm n f_e$

Design requirement: The first torsional frequency must exceed $f_{n1} > 100$ Hz to avoid the GPU PSD high-energy band. Alternatively, virtual torsional damping via SiC DC/DC control can be employed.

4 Stress Analysis and Load Decomposition

4.1 Steady-State Centrifugal Stress

At rated speed (18,000 rpm, $\omega = 1885$ rad/s), centrifugal stress dominates. For a uniform solid disk:

$$\sigma_r(r) = \frac{3+\nu}{8}\rho\omega^2(R^2 - r^2) \tag{10}$$

$$\sigma_\theta(r) = \frac{3+\nu}{8}\rho\omega^2\left(R^2 - \frac{1+3\nu}{3+\nu}r^2\right) \tag{11}$$

Maximum stress at the rotor center:

$$\sigma_{max} = \sigma_r(0) = \sigma_\theta(0) = \frac{3+\nu}{8}\rho\omega^2 R^2 \tag{12}$$

Substituting $R = 0.215$ m for the $$430 mm rotor:

$$\sigma_{max,centrifugal} \approx \frac{3+0.3}{8} \times 7850 \times (1885)^2 \times 0.215^2 \approx 532 \text{ MPa} \tag{13}$$

4.2 Steady Stress Distribution

Location	Stress (MPa)	Source
Shaft center	500–657	FE (Honghui)
Disk root ($R = 0.35$ m)	340–400	FE
Shaft shoulder transition	260–320	FE
Bearing fit surface	120–200	FE

4.3 Dynamic Stress Component

Dynamic stress arises from electromagnetic torque fluctuations causing torsional shear stress:

$$\tau_{xy}(t,x) = \frac{T(t)\cdot r}{J(x)} \tag{14}$$

The von Mises equivalent alternating stress amplitude:

$$\sigma_{vM,alt}(x) = \sqrt{3}\cdot\tau_{xy,alt}(x) \tag{15}$$

4.3.1 Correction Factors

Critical stress concentration locations require correction:

- **Fatigue notch factor**: $K_f = 1 + q(K_t - 1)$, where $q \approx 0.85$–0.95
- **Size factor**: $\varepsilon \approx 0.6$–0.75 for sections > 200 mm
- **Surface factor**: $\beta \approx 0.85$–0.95 for ground surfaces

Effective dynamic stress amplitude:

$$\sigma_{a,eff} = \frac{K_f}{\varepsilon\beta}\cdot\sigma_{a,nominal} \tag{16}$$

4.3.2 Numerical Example (500 kW Rotor)

Assuming the torque fluctuation $T_{alt} \approx 2\%$ of rated torque ($\Delta T \approx 16$ N·m):

Parameter	Value	Unit
Torque fluctuation ΔT	16	N·m
Shear stress amplitude τ_{alt}	6.6	MPa

von Mises stress σ_{vM}	11.5	MPa
$K_f/(\varepsilon\beta)$	3.0	—
Effective stress $\boldsymbol{\sigma_{a,eff}}$	**34.7**	**MPa**
Safety margin vs. S_e (550 MPa)	**15.8×**	—

5 Fatigue Life Assessment

5.1 Material S-N Curve

The 25Cr2Ni4MoV S-N curve follows the Basquin equation:

$$\sigma_a(N_f) = \sigma_f{}' \cdot (2N_f)^b \tag{17}$$

Key fatigue parameters (quenched + tempered):

Parameter	Value	Source
$\sigma_f{}'$	3,890 MPa	$3.7\sigma_b$
b	−0.085	High-strength alloy
Fatigue limit S_e $(N = 10^7)$	550 MPa	Rotating bending test
Torsional fatigue limit τ_e	318 MPa	$\tau_e \approx 0.577S_e$

5.2 Static Strength vs. Fatigue Life

A critical distinction must be made:

Criterion	Meaning	Condition
Static strength	Max load without failure	$\sigma_{max} < \sigma_y/S$
Fatigue life	Cyclic damage accumulation	$D = \sum n_i/N_{fi} < 1$
Fracture safety	Crack stability	$K_{max} < K_{IC}$

The AI data center flywheel operates at stress levels well below static limits ($< 0.6\sigma_y$); **fatigue is the governing failure mode**.

5.3 PSD-Based Random Fatigue

5.3.1 Stress PSD Derivation

From the electromagnetic torque PSD $S_{Te}(f)$:

$$S_\sigma(f, x) = |H_{TV}(f, x)|^2 \cdot S_{Te}(f) \tag{18}$$

where H_{TV} is the torsional vibration transfer function:

$$H_{TV}(f, x) = \sum_{r=1}^{N} \frac{\Phi^{(r)}(x) \cdot \Phi^{(r)}(x_e)}{1 - \left(f/f_n^{(r)}\right)^2 + i \cdot 2\zeta_r \left(f/f_n^{(r)}\right)} \cdot \frac{r_x}{J(x)} \tag{19}$$

5.3.2 Spectral Moments

Define the k-th spectral moment:

$$m_k = \int_0^\infty f^k \cdot S_\sigma(f)\, df \tag{20}$$

Key moments: - $m_0 = \sigma_{rms}^2$ (total variance) - $E[0] = \sqrt{m_2/m_0}$ (zero-crossing rate) - $E[P] = \sqrt{m_4/m_2}$ (peak rate)

5.3.3 Irregularity Factor

$$\gamma = \frac{m_2}{\sqrt{m_0 \cdot m_4}}, \quad 0 \le \gamma \le 1 \tag{21}$$

For AI data center conditions: $\gamma \approx 0.3$–0.6 (broadband random).

5.3.4 Dirlik Rainflow Method

The broadband stress amplitude distribution is directly approximated by the Dirlik formula [2]“:

$$p_{RF}(S) = \frac{1}{\sqrt{m_0}} \cdot \frac{D_1}{Q} e^{-Z/Q} + \frac{D_2 Z}{R^2} e^{-Z^2/(2R^2)} + D_3 Z e^{-Z^2/2} \tag{22}$$

where $Z = S/\sqrt{m_0}$, with coefficients D_1, D_2, D_3, Q, R derived from the spectral moments.

5.3.5 Miner Cumulative Damage

Total damage over operating time T:

$$D = E[P] \cdot T \int_0^\infty \frac{p_{RF}(S)}{N_f(S)} \, dS \tag{23}$$

Fatigue limit: For $D < D_{cr}$ (typically 1.0, conservatively 0.5–0.8).

5.4 Mean Stress Correction

The steady centrifugal stress provides a non-zero mean stress. Using the **Gerber correction** (recommended for ductile steels):

$$\frac{\sigma_a}{S_e} + \left(\frac{\sigma_m}{\sigma_b}\right)^2 = 1 \tag{24}$$

With $\sigma_m \approx 500$ MPa and $\sigma_b = 1,050$ MPa:

Correction	Equivalent σ_a	Factor
None	34.7 MPa	1.0×
Goodman	51.4 MPa	1.48×
Gerber	**41.6 MPa**	**1.20×**

Safety margin vs. S_e (550 MPa) with Gerber correction: **13.2×**.

Conclusion: The rotor has **infinite fatigue initiation life** under AI data center conditions.

6 Crack Propagation Analysis

6.1 Fracture Mechanics Foundation

Assume manufacturing NDT (UT/MPI) detects defects less than inspection limits:

- Internal forging defects: $a_0 \approx 0.5$–1.0 mm (UT limit)
- Surface defects: $a_0 \approx 0.1$–0.3 mm (MPI limit)

6.2 Paris Law

Crack growth under cyclic loading follows the Paris-Erdogan relation:

$$\frac{da}{dN} = C(\Delta K)^m \tag{25}$$

The stress intensity factor range:

$$\Delta K = Y(a) \cdot \Delta\sigma \cdot \sqrt{\pi a} \tag{26}$$

For 25Cr2Ni4MoV (quenched + tempered, $K_{IC} \approx 130$ MPa·m′$^{1/2}$) [3]“:

$$\frac{da}{dN} = 2.1 \times 10^{-12} \cdot (\Delta K)^{3.2} \quad \left(\text{m/cycle, MPa·m}^{1/2}\right) \tag{27}$$

6.3 Mode II/III Dominance

Under torsional excitation, the cyclic crack driving force is:

$$\Delta K_{II/III} = Y_{II/III} \cdot \Delta\tau_{xy} \cdot \sqrt{\pi a} \tag{28}$$

where $\Delta\tau_{xy}$ is determined from the torque PSD rms amplitude.

6.4 Crack Growth Life

Integrating the Paris law:

$$N_p = \int_{a_0}^{a_c} \frac{da}{C(\Delta K(a))^m} \tag{29}$$

For $m \neq 2$:

$$N_p = \frac{2}{(m-2)C(\Delta\sigma)^m Y^m \pi^{m/2}} \left(a_0^{1-m/2} - a_c^{1-m/2}\right) \tag{30}$$

Critical crack size for unstable propagation:

$$a_c = \frac{1}{\pi}\left(\frac{K_{IC}}{Y \cdot \sigma_{max}}\right)^2 \approx 30.4 \ \text{ mm} \tag{31}$$

6.5 ΔK Threshold Analysis

Case	Shaft Diameter	$\Delta\tau_{xy}$	a_0	ΔK	Threshold	Status
500 kW	50 mm	6.6 MPa	1.0 mm	1.38	5–8	≪ threshold
3200 kW	50 mm	2.1 MPa	1.0 mm	0.44	5–8	≪ threshold
3200 kW	100 mm	0.13 MPa	1.0 mm	0.03	5–8	≪ threshold
3200 kW	100 mm	0.13 MPa	0.5 mm	0.02	5–8	≪ threshold

All cases have $\boldsymbol{\Delta K \ll \Delta K_{th}}$, indicating **infinite crack growth life**.

6.6 Random Load Spectral Integration

For broadband random loading, the spectral form is:

$$\overline{\frac{da}{dt}} = E[P] \cdot C \cdot \int_0^\infty (\Delta K)^m \cdot p_{RF}(S)\, dS \tag{32}$$

Substituting $\Delta\sigma_{rms}$ would underestimate crack growth rate by 3–10× under AI data center conditions.

7 AI Data Center vs. Traditional UPS Comparison

7.1 Load Characteristic Comparison

Feature	Traditional UPS	AI Data Center	Difference
Excitation source	Grid fluctuation (0.01–0.5 Hz)	GPU power noise (10 Hz–1 kHz)	2–5 orders
Annual cycles	10^5–10^6	10^8–10^9	10^2–10^3 ×

Bandwidth	Narrowband	Broadband continuum	Fundamental
Dynamic stress ratio	$<5\%$ centrifugal	5%–15% centrifugal	3–5×
Resonance risk	Single-point crossing	Persistent broadband	Fundamental
Non-Gaussianity	Near-Gaussian	Crest factor 4–6	\$2\$

7.2 Lifetime Assessment Comparison

Assessment	Traditional UPS	AI Data Center	Conclusion
Static margin	\$2.5\$	\$2.5\$	Equivalent
HCF life	$>10^9$ (infinite)	10^8–10^9	Reduces 1–2 orders
Crack growth	$>10^{11}$ (negligible)	10^7–10^8 cyc.	Needs monitoring
Dominant failure	Overload/bearing	HCF + crack growth	Fundamental shift

7.3 Power Scalability

The analysis was extended to 3200 kW by mechanical self-scaling:

[For a torque increase of 6.4×, shaft diameter scales by $\sqrt[3]{T}$, or approximately 1.9×. The resulting dynamic stress increases by only 1.1×, remaining far below the fatigue limit.]

Key insight: The rotor fatigue life conclusion is **scale-invariant** across 500 kW–3200 kW due to the cube-root diameter scaling of torsional stress.

8 Conclusion

8.1 Core Findings

1. **Infinite fatigue initiation life**: The Gerber-corrected equivalent stress amplitude (41.6 MPa) is far below the fatigue limit (550 MPa), yielding a safety margin of 13.2×.
2. **Infinite crack growth life**: All examined cases show $\Delta K \ll \Delta K_{th}$ (1.38 MPa·m$'^{1/2}$ vs. threshold 5–8 MPa·m$'^{1/2}$), providing no crack propagation driving force under standard NDT conditions.
3. **Scale invariance**: The infinite fatigue life conclusion holds for both 500 kW and 3200 kW systems due to mechanical self-scaling properties.
4. **Actual life-limiting components**: Bearing system life (15–20 years) and power electronics service life (10–15 years) determine the practical system lifetime.

8.2 Recommended Design Criteria

Criterion	Requirement	Basis
1st torsional frequency	$f_{n1} > 100$ Hz	Avoid GPU PSD high-energy band
Initial defect limit	$a_0 \leq 0.5$ mm	Standard NDT
Safety factor	$D < 0.5$	Conservative Miner criterion
Crack growth SF	$2\times$	Load statistical uncertainty

8.3 Online Monitoring

- **Shaft torsional vibration**: Fiber-optic sensor, 1 kHz sampling
- **Radial displacement**: Laser probes for early crack detection
- **Bus power monitoring**: 2 MHz sampling for torque PSD estimation

- **Acoustic emission**: 150 kHz center frequency for crack growth events

8.4 Design Optimization

1. Material: 25Cr2Ni4MoV with surface nitriding (S_e to 580–620 MPa)
2. Resonance: Increase shaft diameter to push $f_{n1} > 150$ Hz
3. Active damping: Virtual torsional damper via SiC DC/DC control
4. Input filtering: SST DC-link LC filter with $f_c \approx 100$ Hz

9 APPENDIX A: 3200 KW SYSTEM VERIFICATION

9.1 Scaling Analysis

The 3200 kW system verification uses the same 25Cr2Ni4MoV rotor material but with adjusted geometry for the higher power rating.

9.1.1 Torque Scaling

$$\frac{T_{3200}}{T_{500}} = \frac{3200}{500} = 6.4 \tag{33}$$

9.1.2 Shaft Diameter Scaling (Stress-Limited)

For constant torsional stress:

$$\frac{D_{3200}}{D_{500}} = \sqrt[3]{\frac{T_{3200}}{T_{500}}} = \sqrt[3]{6.4} \approx 1.86 \tag{34}$$

9.1.3 Resulting Dynamic Stress

For 50 mm shaft (unchanged diameter, magnetic bearings):

$$\Delta\tau_{xy,3200,50mm} = 6.6 \times \frac{500}{3200} \times \frac{18,000}{25,000} \approx 2.1 \quad \text{MPa} \tag{35}$$

For 100 mm shaft (scaled):

$$\Delta\tau_{xy,3200,100mm} = 2.1 \times \left(\frac{50}{100}\right)^3 \approx 0.13 \quad \text{MPa} \tag{36}$$

9.1.4 ΔK Verification

Shaft (mm)	$\Delta\tau_{xy}$ (MPa)	a_0 (mm)	ΔK (MPa · m$'^{1/2}$)	Threshold	Status
50	2.1	1.0	0.44	5–8	Safe
100	0.13	1.0	0.03	5–8	Safe
100	0.13	0.5	0.02	5–8	Safe

9.1.5 Centrifugal Stress at Higher Speed

For 3200 kW, higher speed (25,000 rpm) may be used:

$$\sigma_{cent,max,3200} = 532 \times \left(\frac{25,000}{18,000}\right)^2 \approx 1026 \quad \text{MPa} \tag{37}$$

This approaches yield strength (1,050 MPa). **Design recommendation**: If $n_{max} > 22,000$ rpm, use larger shaft diameter or material upgrade.

9.2 Appendix B: Key Notation

Symbol	Description	Unit
D	Rotor diameter	mm
E	Young's modulus	GPa
K_{IC}	Fracture toughness	$\mathrm{MPa} \cdot \mathrm{m}'^{1/2}$
ΔK	SIF range	$\mathrm{MPa} \cdot \mathrm{m}'^{1/2}$
ΔK_{th}	Threshold SIF range	$\mathrm{MPa} \cdot \mathrm{m}'^{1/2}$
S_e	Fatigue limit	MPa
σ_a	Alternating stress	MPa
σ_m	Mean stress	MPa
σ_y	Yield strength	MPa
σ_b	Ultimate strength	MPa
τ_{xy}	Shear stress	MPa
f_{sw}	SiC switching frequency	kHz
ω	Angular velocity	rad/s
$\Gamma(f)$	Coherence function	—

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